7 (a) (i)	*either* helium nucleus *or* contains 2 protons and 2 neutrons ………………………………….	B1	[1]
(ii)	e.g. range is a few cm in air/sheet of thin paper speed up to 0.1 *c* causes dense ionisation in air positively charged or deflected in magnetic or electric fields (*any two, 1 each to max 2*) ………………………………………………..	B2	[2]
(b) (i)	${}^{4}_{2}\alpha$ …………………………………………………………………	B1	
	either ${}^{1}_{1}$p *or* ${}^{1}_{1}$H…………………………………………………..	B1	[2]
(ii) 1	initially, α-particle must have some kinetic energy …………………..	B1	[1]
(ii) 2	1.1 MeV = $1.1 \times 1.6 \times 10^{-13} = 1.76 \times 10^{-13}$ J ………………………….	C1	
	$E_K = ½mv^2$ …………………………………………………………..	C1	
	$1.76 \times 10^{-13} = ½ \times 4 \times 1.66 \times 10^{-27} \times v^2$ ………………………………	C1	
	$v = 7.3 \times 10^6$ m s^{-1} ………………………………………………….....	A1	[4]
	use of 1.67×10^{-27} kg for mass is a maximum of 3/4		